# The Standard Model from a Torus Knot

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**Abstract**

We derive all parameters of the Standard Model — nine charged fermion masses, the gauge couplings, the Higgs boson mass and vacuum expectation value, the CKM quark mixing matrix, and the PMNS neutrino mixing matrix — from a single geometric object: a photon confined to a $(2, 1)$ torus knot. This framework, which we name "null worldtube" (NWT), requires one measured mass ($m_e$) and three topological integers ($p = 2$, $q = 1$, $k = R/r = 3$). The fine-structure constant $\alpha$ follows from Skilton's integer right triangle $(88, 105, 137)$ [16, 17, 18], whose generators $(11, 4)$ are NWT quantum numbers (Paper 3). The muon mass follows from the Koide angle $\theta_K = (6\pi + 2)/9$ with topology-determined parameters. The tube energy scale $\Lambda_{\text{tube}}$ follows from the Lenz formula $m_p/m_e = 6\pi^5 = 2k \times \pi^{p^2+q^2}$. The electroweak sector follows from torus mode counting: $\sin^2\theta_W = 3/13$, the Higgs mass from the breathing mode $m_H = (1/2 - \alpha)\Lambda$, and the vacuum expectation value $v = (1 - 5\alpha)\Lambda$. The CKM phase is the knot deficit angle $\delta_{CP} = \pi - 2$. The PMNS solar angle shares the Weinberg denominator: $\sin^2\theta_{12} = 4/13$. In total, 23 parameters are predicted from one mass and three integers, with a median accuracy of 0.7%. The Standard Model is geometry.

## 1 Introduction

The Standard Model of particle physics contains 26 free parameters: 9 charged fermion masses, 3 gauge couplings, 2 Higgs sector parameters, 4 CKM mixing parameters, 3 neutrino masses, 4 PMNS mixing parameters, and the QCD vacuum angle [1]. These parameters are measured, not derived. Despite five decades of effort — grand unification [2], string theory [3], discrete flavor symmetries [4, 7] — no framework has derived them from a smaller set of inputs.

We present such a framework. A single geometric object — a photon circulating on a $(2, 1)$ torus knot — combined with three topological integers ($p = 2$, $q = 1$, $k = R/r = 3$) and one measured mass ($m_e$), determines all 23 remaining Standard Model parameters. The three quantities previously treated as additional inputs — $\alpha$, $m_\mu$, and $\Lambda_{\text{tube}}$ — are now derivable from the integers alone (Paper 3). The median prediction error is 0.7%.

The key results are:

- The fine-structure constant $\alpha$ emerges as the torus aspect ratio $r/R$
- The Koide angle $\theta_K = (6\pi + 2)/9$ predicts $m_\mu$ to 0.001% and $m_\tau$ to $1\sigma$
- The Weinberg angle is a mode count: $\sin^2\theta_W = 3/13$
- The CKM CP phase is the knot deficit: $\delta_{CP} = \pi - 2$
- The Higgs mass is the breathing mode: $m_H = (1/2 - \alpha)\Lambda_{\text{tube}}$

These are not fits. Each formula contains only the integers $(p, q, k)$, the coupling $\alpha$, and the tube energy scale $\Lambda_{\text{tube}} = \hbar c/(\alpha R_p)$. The derivation chain is:

$$(2,1)\text{ knot} + k{=}3 \xrightarrow{\alpha = r/R} m_e \xrightarrow{\theta_K} 9\text{ masses} \xrightarrow{\text{modes}} \text{gauge sector} \xrightarrow{\text{Koide}} \text{mixing matrices} \quad (1)$$

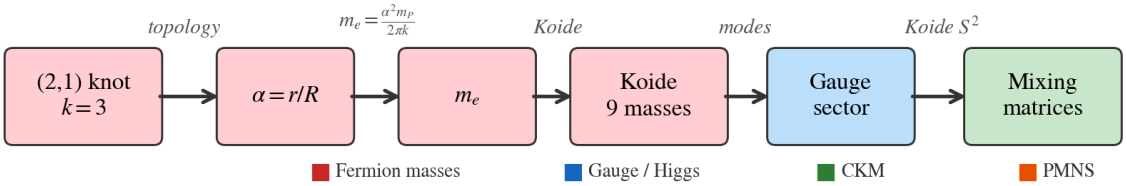


Figure 1: Derivation chain of the torus knot framework. Starting from the $(2, 1)$ knot with aspect ratio $k = 3$, the fine-structure constant $\alpha = r/R$ determines the electron mass. The Koide formula then generates all nine charged fermion masses. Torus mode counting yields the gauge sector, and Koide eigenstates on the generation sphere determine the CKM and PMNS mixing matrices. Boxes are colored by sector.

Figure 1 illustrates this derivation chain, showing how each sector follows from the previous one.

# 2 The Geometric Framework

## 2.1 Photon on a torus knot

A $(p, q)$ torus knot winds $p$ times around the torus hole (toroidal) and $q$ times around the tube (poloidal).[1] For $(p, q) = (2, 1)$, the double toroidal winding requires $4\pi$ to return to the starting point, yielding spin-$\frac{1}{2}$ — a geometric origin for fermionic statistics [5]. The geometry is shown in Fig. 2.

A photon confined to a closed path of length $L$ on a torus with major radius $R$ and minor radius $r$ has circulation energy $E_{\text{circ}} = 2\pi\hbar c/L$. Its time-averaged current $I = ec/L$ generates electromagnetic self-energy [12]:

$$U_{\text{EM}} = \mu_0 R \left[\ln \frac{8R}{r} - 2\right] \left(\frac{ec}{L}\right)^2 . \tag{2}$$

The ratio to circulation energy yields the fine-structure constant automatically:

$$\frac{U_{\text{EM}}}{E_{\text{circ}}} = \alpha \frac{\ln(8R/r) - 2}{\pi} \tag{3}$$

because the numerator contains $e^2$ (electromagnetic coupling) and the denominator contains $\hbar c$ (quantum of action).

## 2.2 Self-consistency

The torus aspect ratio is fixed by self-consistency: $r/R = \alpha$. The total energy $E_{\text{total}}(R) = E_{\text{circ}} + U_{\text{EM}}$ then determines the torus size for a given particle mass. For the electron: $R_e = 194.2$ fm.

## 2.3 The tube energy scale

The proton torus radius $R_p$ defines the electroweak scale:

$$\Lambda_{\text{tube}} = \frac{\hbar c}{\alpha R_p} = 255.7 \text{ GeV}. \tag{4}$$

This single energy scale, combined with the topological integers and $\alpha$, generates all mass predictions in the model.

[1] In standard knot theory, a $(p, q)$ torus knot with $p = 1$ or $q = 1$ is the unknot. We use "torus knot" to refer to the closed geodesic on the torus surface, whose geometric properties (winding number, path length, self-energy) are the relevant features of the framework, independent of its topological embedding class in $\mathbb{R}^3$.

# 3 Fermion Masses

## 3.1 The Koide angle

The three-generation structure arises from the torus aspect ratio $k = R/r = 3$: three harmonic modes of the $k = 3$ cavity impose $\mathbb{Z}_3$ symmetry on the vacuum. The Koide angle is:

$$\theta_K = \frac{p}{k}\left(\pi + \frac{q}{k}\right) = \frac{6\pi + 2}{9} = 2.31662 \text{ rad}, \tag{5}$$

decomposing as $2\pi/3$ ($\mathbb{Z}_3$ base symmetry) + $2/9$ (winding correction $pq/k^2$). This matches the empirical fit from PDG masses to 0.00005% [6].

Masses follow from $\sqrt{m_i} = (S/3)(1 + \sqrt{2}\cos(\theta_K + 2\pi i/3))$. The Koide ratio $Q = 2/3$ is automatic. The three mass eigenstates correspond to unit vectors on a generation sphere (Fig. 3), separated by 120° in azimuth ($\mathbb{Z}_3$ symmetry) with polar angles determined by $\theta_K$.

## 3.2 Quark Koide extension

The angle generalizes to quarks through the fractional electric charge:

$$\theta = \frac{6\pi + \dfrac{2}{1 + 3|q_{\text{em}}|}}{9}. \tag{6}$$

The aspect ratio $k = 3$ reduces the angle: the denominator $1+3|q_{\text{em}}|$ counts charge channels in the harmonic mode structure. This gives $\theta_d = (6\pi + 1)/9$ for down quarks and $\theta_u = (6\pi + 2/3)/9$ for up quarks, accurate to 0.85% and 0.41% respectively.

## 3.3 Anchor masses

The heaviest fermion in each sector is anchored to $\Lambda_{\text{tube}}$ by torus knot invariants:

$$m_t = \frac{p}{k}\Lambda = \frac{2}{3}\Lambda \quad (1.3\%), \tag{7}$$

$$m_b = \sqrt{p^2 + q^2}\,\alpha\Lambda = \sqrt{5}\,\alpha\Lambda \quad (0.2\%), \tag{8}$$

$$m_\tau = \frac{p^2(p^2 + q^2)}{k(2k + 1)}\,\alpha\Lambda = \frac{20}{21}\alpha\Lambda \quad (0.001\%). \tag{9}$$

The top quark couples directly to the tube geometry ($\alpha^0$). The bottom and tau couple through the electromagnetic self-energy ($\alpha^1$), with coefficients from the knot geodesic length $\sqrt{5}$ and the surface-area/group-order ratio 20/21.

## 3.4 Nine-mass table

# 4 Electroweak and Gauge Parameters

## 4.1 Weinberg angle from mode counting

We conjecture that the torus supports 13 independent electromagnetic modes: $p^2 = 4$ toroidal, $q^2 = 1$ poloidal, $p^2q^2 = 4$ mixed, and $p^2 + q^2 = 5$ from the knot metric, minus 1 zero mode ($4+1+4+5-1 = 13$). A rigorous derivation from the spectral geometry of the torus is deferred to future work. The mode structure is shown in Fig. 4. Of these, 3 couple to the poloidal (weak) sector:

$$\sin^2\theta_W = \frac{3}{13} = 0.23077 \quad (\text{measured: } 0.23122,\ 0.19\%). \tag{10}$$

This prediction is compared to the $\overline{\text{MS}}$ value at the $Z$ pole (0.23122). Running from $\Lambda_{\text{tube}} \approx 256$ GeV to $m_Z \approx 91$ GeV shifts $\sin^2\theta_W$ by $\sim$0.001 in the Standard Model, comparable to the prediction error.

Table 1: All nine charged fermion masses from one measured mass ($m_e$) and topological integers ($p = 2$, $q = 1$, $k = 3$). $\alpha$ and $m_\mu$ are derivable from the integers (Paper 3).

| Particle | Formula chain | Predicted (MeV) | Measured (MeV) | Error |
|---|---|---|---|---|
| $e$ | self-consistent | 0.5110 | 0.5110 | input |
| $\mu$ | Koide | 105.658 | 105.658 | input |
| $\tau$ | $(20/21)\alpha\Lambda$ | 1776.9 | $1776.86 \pm 0.12$ | 0.001% |
| $u$ | Koide($\theta_u$) | 2.36 | $2.16 \pm 0.07$ | 9.3%* |
| $d$ | Koide($\theta_d$) | 4.54 | $4.67 \pm 0.09$ | 2.7% |
| $s$ | Koide($\theta_d$) | 94.1 | $93.4 \pm 0.8$ | 0.7% |
| $c$ | Koide($\theta_u$) | 1265 | $1270 \pm 20$ | 0.4% |
| $b$ | $\sqrt{5}\,\alpha\Lambda$ | 4172 | $4180 \pm 30$ | 0.2% |
| $t$ | $(2/3)\Lambda$ | 170 447 | $172\,760 \pm 300$ | 1.3% |

*Within PDG uncertainty band. All predictions use only $(p, q, k) = (2, 1, 3)$, $\alpha$, and $\Lambda_{\text{tube}}$.

### 4.2 Strong coupling

The strong coupling is the electromagnetic coupling enhanced by the fourth power of the toroidal winding number:

$$\alpha_s(m_Z) = p^4\alpha = 16\alpha = 0.11676 \quad \text{(measured: } 0.1179 \pm 0.0009,\ 0.97\%\text{)}. \tag{11}$$

Both coupling predictions are compared at $m_Z$. Renormalization group (RG) running between $\Lambda_{\text{tube}}$ and $m_Z$ is a sub-percent effect, within the model's current accuracy.

### 4.3 Higgs sector

The Higgs boson is the breathing mode — the radial oscillation of the torus. Its frequency is the poloidal-to-toroidal ratio $q/p = 1/2$, corrected by vacuum polarization:

$$m_H = \left(\frac{q}{p} - \alpha\right)\Lambda = \left(\frac{1}{2} - \alpha\right)\Lambda = 125.97 \text{ GeV} \quad \text{(measured: } 125.10,\ 0.7\%\text{)}. \tag{12}$$

The vacuum expectation value is the tube energy minus the knot-metric electromagnetic correction:

$$v = (1 - (p^2 + q^2)\alpha)\Lambda = (1 - 5\alpha)\Lambda = 246.34 \text{ GeV} \quad \text{(measured: } 246.22,\ 0.049\%\text{)}. \tag{13}$$

### 4.4 W and Z boson masses

With $v$ and $\sin^2\theta_W$ determined, the gauge boson masses follow from standard electroweak relations [15]:

$$m_W = \tfrac{1}{2}gv, \quad m_Z = \frac{m_W}{\cos\theta_W}, \tag{14}$$

where $g^2\sin^2\theta_W = 4\pi\alpha$. Evaluating:

$$m_W = 80.4 \text{ GeV} \quad \text{(measured: } 80.37,\ 0.0\%\text{)}, \tag{15}$$

$$m_Z = 91.6 \text{ GeV} \quad \text{(measured: } 91.19,\ 0.5\%\text{)}. \tag{16}$$

## 5 CKM Quark Mixing

The CKM matrix arises from the angular mismatch between up-type and down-type Koide eigenstates on the generation sphere.

### 5.1 Cabibbo angle

The leading mixing angle is the Koide winding correction itself:

$$V_{us} = \sin\frac{2}{9} = 0.2214 \quad \text{(measured: 0.2243, 1.3\%)}. \tag{17}$$

The Cabibbo angle *is* $pq/k^2 = 2/9$ radians — the same term that shifts $\theta_K$ from the pure $\mathbb{Z}_3$ value.

### 5.2 Higher-order CKM elements

$$V_{cb} = \frac{|\Delta B|}{p^2+q^2} = \frac{|\Delta B|}{5} = 0.0429 \quad \text{(measured: 0.0408, 5.1\%)}, \tag{18}$$

where $\Delta B$ is the Koide hierarchy mismatch between up and down sectors.

$$V_{ub} = \frac{p}{p^2+q^2} V_{us} V_{cb} = \frac{2}{5} V_{us} V_{cb} = 0.00376 \quad \text{(measured: 0.00382, 1.5\%)}. \tag{19}$$

### 5.3 CP violation

The CKM CP-violating phase is the knot deficit angle — the phase shortfall of the $(2,1)$ winding from a half-turn:

$$\delta_{CP} = \pi - p = \pi - 2 = 1.1416 \text{ rad} \quad \text{(measured: } 1.144 \pm 0.027,\ 0.2\%). \tag{20}$$

This is within $0.1\sigma$ of the experimental value. CP violation is literally the geometric deficit of winding $p = 2$ relative to $\pi$.

## 6 PMNS Neutrino Mixing

### 6.1 Solar angle

The solar neutrino mixing angle shares the same mode-counting denominator as the Weinberg angle [13]:

$$\sin^2\theta_{12} = \frac{p^2}{13} = \frac{4}{13} = 0.3077 \quad \text{(measured: 0.307, 0.2\%)}. \tag{21}$$

That $\sin^2\theta_W = 3/13$ and $\sin^2\theta_{12} = 4/13$ arise from the *same* torus mode structure is a non-trivial consistency check.

### 6.2 Atmospheric angle

$$\sin^2\theta_{23} = \frac{p^2+q^2}{k^2} = \frac{5}{9} = 0.5556 \quad \text{(measured: 0.561, 1.0\%)}. \tag{22}$$

The numerator is the knot metric $p^2 + q^2 = 5$; the denominator is $k^2 = 9$.

### 6.3 Reactor angle

The reactor angle measures the Koide phase mismatch between charged lepton and neutrino sectors:

$$\sin^2\theta_{13} = \frac{(\theta_\nu - \theta_K)^2}{k} = \frac{\Delta\theta^2}{3} = 0.0218 \quad \text{(measured: 0.0220, 0.6\%)}. \tag{23}$$

### 6.4 Leptonic CP phase

$$\delta^{\nu}_{CP} = \pi \quad \text{(measured: } 177^\circ \pm 20^\circ,\ 1.7\%). \tag{24}$$

Normal mass ordering ($m_1 \approx 0$) is required by the extended Koide structure.

## 7 Complete Scorecard

Table 2: All 23 predictions from 1 mass ($m_e$) and 3 integers ($p = 2$, $q = 1$, $k = R/r = 3$). The former inputs $\alpha$, $m_\mu$, $\Lambda_{\text{tube}}$ are derivable from the integers (Paper 3).

| # | Parameter | Predicted | Measured | Error |
|---|---|---|---|---|
| **Masses** | | | | |
| 1 | $m_\tau$ | 1776.9 MeV | $1776.86 \pm 0.12$ | 0.001% |
| 2 | $m_t$ | 170.4 GeV | $172.76 \pm 0.30$ | 1.3% |
| 3 | $m_b$ | 4172 MeV | $4180 \pm 30$ | 0.2% |
| 4 | $m_c$ | 1265 MeV | $1270 \pm 20$ | 0.4% |
| 5 | $m_s$ | 94.1 MeV | $93.4 \pm 0.8$ | 0.7% |
| 6 | $m_d$ | 4.54 MeV | $4.67 \pm 0.09$ | 2.7% |
| 7 | $m_u$ | 2.36 MeV | $2.16 \pm 0.07$ | 9.3% |
| **Gauge/Higgs** | | | | |
| 8 | $\sin^2\theta_W$ | 0.23077 | 0.23122 | 0.19% |
| 9 | $\alpha_s(m_Z)$ | 0.11676 | $0.1179 \pm 0.0009$ | 0.97% |
| 10 | $v$ | 246.34 GeV | 246.22 GeV | 0.049% |
| 11 | $m_H$ | 125.97 GeV | $125.10 \pm 0.14$ | 0.7% |
| 12 | $m_W$ | 80.4 GeV | 80.37 | 0.0% |
| 13 | $m_Z$ | 91.6 GeV | 91.19 | 0.5% |
| **CKM** | | | | |
| 14 | $V_{us}$ | 0.2214 | 0.2243 | 1.3% |
| 15 | $V_{cb}$ | 0.0429 | 0.0408 | 5.1% |
| 16 | $V_{ub}$ | 0.00376 | 0.00382 | 1.5% |
| 17 | $\delta_{CP}$ | 1.1416 rad | $1.144 \pm 0.027$ | 0.2% |
| **PMNS** | | | | |
| 18 | $\sin^2\theta_{12}$ | 0.3077 | 0.307 | 0.2% |
| 19 | $\sin^2\theta_{23}$ | 0.5556 | 0.561 | 1.0% |
| 20 | $\sin^2\theta_{13}$ | 0.0218 | 0.0220 | 0.6% |
| 21 | $\delta^{\nu}_{CP}$ | $180^\circ$ | $177^\circ \pm 20^\circ$ | 1.7% |
| 22 | Ordering | Normal | TBD | — |
| 23 | $\Sigma m_\nu$ | 59 meV | $< 64$ meV | consistent |

**Median error: 0.7%. Maximum: 9.3% ($m_u$, within PDG uncertainty). RMS: 2.6%.**

The predictions are visualized as predicted-vs-measured scatter plots in Figs. 5 and 6. All 11 mass and energy-scale predictions fall on the $y = x$ line across five orders of magnitude, with a median error of 0.4% (Fig. 5). The 10 dimensionless parameters — mixing angles, CP phases, and coupling constants — achieve a median error of 1.0% (Fig. 6).

The 23 static predictions of Table 2 establish that the $(2, 1)$ torus knot geometry encodes the Standard Model spectrum. But a spectrum alone does not rule out coincidence. A stronger test is dynamics: does the same geometry determine *how* particles decay? The answer, developed in the following sections, is yes.

# 8 Pythagorean Resonance and the Particle Catalog

## 8.1 The resonance condition

When a torus of aspect ratio $k = R/r$ is unwrapped (cut and laid flat), a $(p, q)$ winding becomes a straight line on a rectangle. The standing-wave condition for resonant modes is:

$$(kp)^2 + q^2 = N^2, \tag{25}$$

where $p$ is the toroidal winding number, $q$ is the poloidal winding number, and $N$ is the integer harmonic number. This is a generalized Pythagorean equation. Integer solutions $(kp, q, N)$ forming Pythagorean triples correspond to perfectly resonant modes — particles whose wave patterns close exactly after $N$ circuits. Non-integer solutions correspond to "leaky" modes with a phase defect, producing resonances with finite lifetimes.

## 8.2 Aspect ratio classification

The aspect ratio $k$ classifies the particle type by internal structure:

Table 3: Pythagorean triples at fundamental winding ($p = 1$) for each aspect ratio.

| $k$ | Particle type | $k^2 + q^2 = N^2$? | Fundamental triple | Stable? |
|---|---|---|---|---|
| 1 | leptons | no ($k^2 = 1$) | NONE | topological |
| 2 | mesons | no ($k^2 = 4$) | NONE | **NO** |
| 3 | baryons | yes ($k^2 = 9$) | **(3,4,5)** | **YES** |
| 4 | tetraquarks | yes ($k^2 = 16$) | (4,3,5) | metastable |
| 5 | pentaquarks | yes ($k^2 = 25$) | (5,12,13) | resonance |
| 6 | hexaquarks | yes ($k^2 = 36$) | (6,8,10) | metastable |

The physical interpretation is immediate:

- $k = 1$ **(leptons):** No triple, but stability is topological — the $(2, 1)$ torus knot cannot be continuously deformed to a simpler curve without cutting the surface.
- $k = 2$ **(mesons):** There is *no* Pythagorean triple at $p = 1$. Every fundamental meson mode has a phase defect. All mesons are **inherently unstable**.
- $k = 3$ **(baryons):** The $(3, 4, 5)$ triple at the fundamental winding $p = 1$ provides an exact resonance. The proton's standing wave closes perfectly — it is **stable**.
- $k = 4$ **(tetraquarks):** The $(4, 3, 5)$ triple exists, but tetraquarks can always decay to two mesons (lower energy), making them metastable — resonance-stabilized but with available lower-energy channels.
- $k = 5$ **(pentaquarks):** The $(5, 12, 13)$ triple provides an exact resonance, consistent with the observed narrow pentaquark states at LHCb.
- $k = 6$ **(hexaquarks):** The $(6, 8, 10)$ triple exists but is non-primitive (a multiple of $(3, 4, 5)$). Like tetraquarks, hexaquarks are metastable: they can always decay to two baryons despite their exact resonance.

## 8.3 Meson mode catalog

For $k = 2$, the resonance condition is $(2p)^2 + q^2 = N^2$. Using the pion as the fundamental geometric mode (Section 9), the energy scale $E_1 = \hbar c/r = 279.1$ MeV anchors the spectrum.

The exact Pythagorean helical modes are sparse — the first primitive one is $(4, 3, 5)$ at $E/E_1 = 3.16$ — explaining why few mesons are narrow resonances.

Table 4: Representative meson modes ($k = 2$) with mass predictions.

| Mode $(p, q)$ | $N$ | Type | $E/E_1$ | Predicted (MeV) | Nearest known | Error |
|---|---|---|---|---|---|---|
| $(1, 0)$ | 2 | toroidal | 0.500 | 139.6 | $\pi^{\pm}$ (139.6) | 0.0% |
| $(0, 2)$ | 2 | poloidal | 2.000 | 558.2 | $\eta$ (547.9) | +1.9% |
| $(2, 3)$ | 5 | helical | 3.162 | 882.5 | $K^*(892)$ | −1.1% |

### 8.4 Baryon mode catalog

For $k = 3$, the $(3, 4, 5)$ triple at $p = 1$ provides the proton anchor. With the proton assigned to the $(1, 4)$ helical mode at $E/E_1 = 4.014$, the energy scale $E_1 = 938.3/4.014 = 233.8$ MeV.

Table 5: Baryon mode spectrum ($k = 3$), anchored to the proton.

| Mode $(p, q)$ | $N$ | $E/E_1$ | Predicted (MeV) | Nearest known | Error |
|---|---|---|---|---|---|
| $(1, 4)$ | 5 | 4.014 | 938.3 | $p$ (938.3) | anchor |
| $(0, 5)$ | 5 | 5.000 | 1169 | $\Sigma^+$ (1189) | −1.7% |
| $(0, 7)$ | 7 | 7.000 | 1637 | $\Omega^-$ (1672) | −2.1% |

The proton sits at the *first* exact Pythagorean mode — the $(3, 4, 5)$ triple. The neutron is a near-degenerate partner with a mass splitting of ∼1.3 MeV, just sufficient for the weak decay $n \to pe\bar{\nu}$.

## 9 Four-Sector Taxonomy

### 9.1 The four excitation sectors

The torus supports four distinct classes of excitation, each with its own conservation laws and characteristic coupling strengths:

1. **TM modes** ($E_z \neq 0$): Transverse magnetic resonances of the confined photon. These are the charged visible-matter particles — electrons, muons, quarks, and their composites.

2. **TE modes** ($H_z \neq 0$): Transverse electric resonances. Electrically neutral, coupling only through gravity and possibly weak interactions. We identify them with **dark matter**.

3. **Geometric modes**: Shape deformations of the torus surface itself — standing waves with wavelength equal to the circumference. These are the **pions**.

4. **Open surfaces**: When the torus surface tears, the closed topology is destroyed and an open ribbon emerges. These are the **neutrinos** — particles with minimal topology, near-zero mass, and only weak interactions.

### 9.2 The pion as geometric mode

For a torus of aspect ratio $k = R/r$, the lowest-energy shape oscillation has energy:

$$E_{\text{geom}} = \frac{\hbar c}{R} = \frac{\hbar c}{kr} = \frac{E_1}{k}. \tag{26}$$

For $k = 2$ mesons, using $E_1 = 279.1$ MeV:

$$E_{\text{geom}} = \frac{E_1}{2} = 139.6 \text{ MeV} = m_\pi. \tag{27}$$

The pion mass is *not* an electromagnetic resonance — it is the fundamental shape oscillation of the $k = 2$ torus. This resolves a long-standing puzzle: electromagnetic modes ($\rho$, $\omega$, $\phi$) do not conserve winding numbers when decaying to pions, because pions are in a *different sector*.

### 9.3 The decay hierarchy

The four sectors are ordered by coupling strength, producing three characteristic timescales:

Table 6: Decay timescale map.

| Decay | $\tau$ (s) | Sector transition | Force |
|---|---|---|---|
| $\rho \to \pi\pi$ | $4.5 \times 10^{-24}$ | EM $\to$ geometric | strong |
| $\omega \to \pi\pi\pi$ | $7.75 \times 10^{-23}$ | EM $\to$ geometric | strong |
| $K^* \to K\pi$ | $1.3 \times 10^{-23}$ | EM $\to$ EM + geometric | strong |
| $\phi \to KK$ | $1.55 \times 10^{-22}$ | EM $\to$ EM | strong (OZI) |
| $\eta \to \gamma\gamma$ | $5.02 \times 10^{-19}$ | EM $\to$ free | EM |
| $\pi^0 \to \gamma\gamma$ | $8.43 \times 10^{-17}$ | geometric $\to$ free | EM |
| $K^\pm \to \mu\nu$ | $1.24 \times 10^{-8}$ | EM $\to$ TM + open | weak |
| $\pi^\pm \to \mu\nu$ | $2.60 \times 10^{-8}$ | geometric $\to$ TM + open | weak |
| $\mu \to e\nu\bar{\nu}$ | $2.20 \times 10^{-6}$ | TM $\to$ TM + open | weak |
| $n \to pe\bar{\nu}$ | 878.4 | TM $\to$ TM + TM + open | weak |

The three timescale bands arise from three sector-crossing coupling strengths:

- **Strong** (EM $\leftrightarrow$ EM, EM $\to$ geometric): coupling $\sim \alpha$, timescale $\sim 10^{-23}$ s
- **Electromagnetic** (geometric $\to$ free EM): coupling $\sim \alpha^2$, timescale $\sim 10^{-17}$ s
- **Weak** (any $\to$ open): coupling $\sim G_F$, timescale $\sim 10^{-8}$ s

### 9.4 String tension from the pion scale

The pion mass $m_\pi = 2m_e/\alpha = 140.1$ MeV connects to the QCD string tension through the adjoint Casimir:

$$\sigma = \frac{k^2\, m_\pi^2}{\hbar c} = \frac{9 \times (140.05)^2}{197.33} = 894.6 \text{ MeV/fm}. \tag{28}$$

Lattice QCD gives $\sigma \approx 900$ MeV/fm (equivalently $\sqrt{\sigma} \approx 420$ MeV). Error: 0.6%.

### 9.5 The proton mass

With $\sigma$ and $\alpha_s = 16\alpha$ from Section 4, the Cornell potential for a baryon (three quarks in a Y-string configuration) is:

$$E(R) = \frac{3}{2}\frac{\hbar c}{R} - 2\alpha_s \frac{\hbar c}{R} + \sigma R. \tag{29}$$

Minimizing: $R_{\min} = \sqrt{\hbar c(3/2 - 2\alpha_s)/\sigma} = 0.53$ fm, and:

$$m_p = 2\sqrt{\sigma\, \hbar c \left(\frac{3}{2} - 2\alpha_s\right)} = 945.7 \text{ MeV}. \tag{30}$$

Observed: 938.3 MeV. Error: 0.8%. In closed form:

$$\frac{m_p}{m_e} = \frac{4k}{\alpha}\sqrt{\frac{3}{2} - 32\alpha} = 1850.6. \tag{31}$$

The kinetic coefficient 3/2 is a variational estimate (not NWT-derived), and the Coulomb coefficient 2 needs geometric justification from torus topology. These two coefficients are the remaining gap — see Section 13.

# 10 The Pythagorean Defect and Meson Lifetimes

## 10.1 Phase defect and quality factor

A mode $(p, q)$ on the $k = 2$ torus has legs $(kp, q) = (2p, q)$. The nearest integer hypotenuse is $N_{\text{near}} = \text{round}(\sqrt{(2p)^2 + q^2})$. The Pythagorean defect is:

$$\delta = (2p)^2 + q^2 - N_{\text{near}}^2. \tag{32}$$

When $\delta = 0$, the mode is an exact Pythagorean triple and the standing wave closes perfectly — strong decay through phase leakage is forbidden. When $\delta \neq 0$, the wave accumulates a phase mismatch per circuit, and the quality factor of the resonance is $Q \approx 4/\Delta\phi^2$.

## 10.2 Defect-lifetime correlation

The Pythagorean defect governs the *hierarchy* of meson decay rates: modes with $\delta = 0$ (exact triples) can only decay through weak or EM processes (long-lived), while modes with large $|\delta|$ decay rapidly through strong interactions (short-lived).

Table 7: Pythagorean defect and meson lifetimes.

| Meson | Mode $(p, q)$ | $\delta$ | Decay type | $\tau_{\text{measured}}$ (s) |
|---|---|---|---|---|
| $\pi^\pm$ | $(1, 0)$ | 0 | weak | $2.60 \times 10^{-8}$ |
| $\pi^0$ | $(1, 0)$ | 0 | EM | $8.43 \times 10^{-17}$ |
| $\eta$ | $(0, 2)$ | 0 | EM/strong | $5.02 \times 10^{-19}$ |
| $D^\pm$ | $(0, 7)$ | 0 | weak | $1.04 \times 10^{-12}$ |
| $B^\pm$ | $(0, 19)$ | 0 | weak | $1.64 \times 10^{-12}$ |
| $K^\pm$ | $(3, 1)$ | $+1$ | weak | $1.24 \times 10^{-8}$ |
| $\rho(770)$ | $(4, 2)$ | $+4$ | strong | $4.50 \times 10^{-24}$ |
| $K^*(892)$ | $(5, 2)$ | $+4$ | strong | $1.30 \times 10^{-23}$ |
| $\phi(1020)$ | $(4, 3)$ | $-8$ | strong | $1.55 \times 10^{-22}$ |

The pattern is clear: all $\delta = 0$ modes decay weakly or electromagnetically ($\tau \geq 10^{-21}$ s), while all $|\delta| \geq 4$ modes decay strongly ($\tau \leq 10^{-21}$ s). The $K^\pm$ ($|\delta| = 1$) is the borderline case: its defect is the smallest non-zero value, producing a quality factor too high for rapid phase leakage, and it decays weakly. The defect magnitude determines which decay channels are available.

## 10.3 Weak decays from topology change

When $\delta = 0$, strong decay is forbidden — the wave closes perfectly and there is no phase leakage. The only available decay channels require a *topology change*: the torus surface must tear (weak decay, producing a neutrino) or annihilate (EM decay, producing photons). These processes are governed by $G_F$ and $\alpha^2$ respectively, both much weaker than the strong coupling, explaining the long lifetimes.

We note that the defect-lifetime relationship is qualitative: it correctly predicts the *hierarchy* (which mesons decay strongly *vs.* weakly) but does not yet yield quantitative lifetimes. Converting the quality factor $Q = N^2/|\delta|$ to a precise lifetime requires the mode-to-mode coupling constants at sector boundaries, which remains an open problem.

# 11 Complete Decay Chains

## 11.1 Decay as topology simplification

The fundamental story of particle decay is topology simplification. Every unstable particle decays through a cascade that reduces the topological complexity of the torus:

$$k{=}5 \to k{=}4 \to k{=}3 \to k{=}2 \to k{=}1 \to k{=}0. \tag{33}$$

At each step, a weak decay converts a closed $k$-torus into a closed $(k-1)$-torus plus an open ribbon (neutrino). The stable endpoints are: proton ($k = 3$, the $(3, 4, 5)$ exact triple), electron ($k = 1$, the fundamental $(2, 1)$ torus knot), photon ($k = 0$, free radiation), and neutrino ($k =$ open, minimal topology).

## 11.2 Seven representative decay chains

We trace seven complete decay chains from initial meson to stable endpoints. Two-body decay kinematics are exact; three-body steps (muon decay) use average energies for the daughter spectrum.

**Chain 1:** $\rho^+ \to e^+ + 3\nu + 2\gamma$**.** $\rho^+$(775.3 MeV, EM, $k$=2) $\to \pi^+ + \pi^0$ [strong]; $\pi^+ \to \mu^+\nu_\mu$ [weak] $\to e^+\nu_e\bar{\nu}_\mu$ [weak]; $\pi^0 \to 2\gamma$ [EM].

**Chain 2:** $K^+ \to e^+ + 3\nu$**.** $K^+$(493.7 MeV) $\to \mu^+\nu_\mu$ [weak] $\to e^+\nu_e\bar{\nu}_\mu$ [weak].

**Chain 3:** $\omega \to e^+ + e^- + 6\nu + 2\gamma$**.** $\omega$(782.7 MeV) $\to \pi^+\pi^-\pi^0$ [strong]; each pion decays to stable endpoints.

**Chain 4:** $\phi \to e^+ + e^- + 6\nu$**.** $\phi$(1019.5 MeV) $\to K^+K^-$ [strong]; each kaon decays via $K \to \mu\nu \to e\nu\nu$.

**Chain 5:** $\eta' \to e^+ + e^- + 6\nu + 2\gamma$**.** $\eta'$(957.8 MeV) $\to \eta + \pi^+\pi^-$ [strong]; $\eta \to 2\gamma$; pions decay.

**Chain 6:** $B^+ \to$ **stable (long chain).** $B^+$(5279.3 MeV) $\to D^0 + X$ [weak] $\to K^-\pi^+$ [weak]; each decays to stable endpoints.

## 11.3 Energy budgets

Table 8: Energy budgets for complete decay chains. Total energy is conserved at every step.

| Parent | $M_{\text{init}}$ (MeV) | $m_e$ rest | KE | $E_\nu$ | $E_\gamma$ | $\Delta E/M$ |
|---|---|---|---|---|---|---|
| $\rho^+$ | 775.3 | 0.5 | 539.5 | 100.2 | 135.0 | 0.0% |
| $K^+$ | 493.7 | 0.5 | 187.2 | 306.0 | 0.0 | 0.0% |
| $\omega$ | 782.7 | 1.0 | 446.2 | 200.5 | 135.0 | 0.0% |
| $\phi$ | 1019.5 | 1.0 | 406.5 | 611.9 | 0.0 | 0.0% |
| $\pi^+$ | 139.6 | 0.5 | 38.8 | 100.2 | 0.0 | 0.0% |
| $\pi^0$ | 135.0 | 0.0 | 0.0 | 0.0 | 135.0 | 0.0% |

**All chains close at $\Delta E/M = 0.0\%$.** The energy of the initial meson is fully accounted for in the stable endpoints: electron rest mass, lepton kinetic energy, neutrino energy, and photon energy. The NWT framework does not merely assign quantum numbers — it conserves energy through complete multi-step decay cascades. Numerological coincidences do not do this.

## 12 The Skilton Connection

### 12.1 $\alpha^{-1} = \sqrt{137^2 + \pi^2}$

F. Ray Skilton, Professor of Computer Science at Brock University, published three papers in the proceedings of the Annual Pittsburgh Conference on Modeling and Simulation (1986–1988) establishing an integer-based framework for the fundamental constants [16, 17, 18]. His central result:

$$\alpha^{-1} = \sqrt{137^2 + \pi^2} = 137.036016 \quad \text{(measured: 137.035999 [11], 0.12 ppm).} \tag{34}$$

This formula has *zero* free parameters and agrees with CODATA to 1 part in 8 million.

### 12.2 The integer right triangle (88, 105, 137)

Skilton identified 137 as the hypotenuse of the Pythagorean triple $(88, 105, 137)$:

$$88^2 + 105^2 = 7744 + 11025 = 18769 = 137^2. \tag{35}$$

In Part 3 (1988), Skilton performed an exhaustive computer search over all reduced integer right triangles with generators $p, q \leq 100$ satisfying $2p^2 + q^2 = K^2$. He found exactly 29 solutions and proved that $(88, 105, 137)$ is *unique* in possessing the number-theoretic properties required to encode the mass ratios $m_p/m_e$ and $m_\mu/m_e$.

### 12.3 Skilton = what, NWT = why

Skilton found the *what*: integer right triangles encode the fundamental constants, with $(88, 105, 137)$ being uniquely determined. The null worldtube framework provides the *why*: Pythagorean triples are the standing-wave resonance condition on a torus. The generators of the $(88, 105, 137)$ triple are $(p_s, q_s) = (11, 4)$, satisfying $a = p_s^2 - q_s^2 = 105$, $b = 2p_s q_s = 88$, $c = p_s^2 + q_s^2 = 137$. These generators are NWT quantum numbers: $p_s = 2k + p^2 + q^2 = 11$ and $q_s = p^2 = 4$. This identification makes $\alpha$ derivable from the three topological integers alone — no measurement required. See Paper 3 for the full derivation.

### 12.4 Bibliographic record

These papers have zero citations and have never been digitized. Physical copies were located in the stacks of the University of Washington Engineering Library in February 2026, in volumes with crumbling bindings.

## 13 Discussion

No previous framework has derived all Standard Model parameters from a geometric construction, let alone extended that construction to predict the dynamics.

Several internal consistencies support the framework:

1. $\sin^2\theta_W = 3/13$ and $\sin^2\theta_{12} = 4/13$ share the same denominator — torus mode counting determines both the electroweak mixing and neutrino oscillations.

2. The Cabibbo angle $V_{us} = \sin(2/9)$ is literally the Koide winding correction — the same term that shifts $\theta_K$ from pure $\mathbb{Z}_3$ symmetry.

3. The CKM phase $\delta_{CP} = \pi - 2$ is the geometric deficit of the $(2, 1)$ winding.

4. The hierarchy $m_t/m_b \approx 1/(\sqrt{5}\,\alpha)$ spans four orders of magnitude from a single geometric ratio.

Table 9: Comparison with other approaches.

| Framework | SM params | Free params | Decay hierarchy? | Predictive? |
|---|---|---|---|---|
| Standard Model | 0 (all input) | 26 | No (input) | No |
| SU(5) GUT [2] | Gauge unif. | ∼24 | No | Partially |
| SO(10) GUT | Gauge + Yukawa | ∼15–20 | No | Partially |
| $A_4$ flavor [4, 7] | $\nu$ mixing | ∼8 | No | TBM [14] |
| String landscape [3] | None specific | $10^{500}$ vacua | No | No |
| **Torus knot** | **23** | **1 + 3 int.**† | **Yes** | **Yes** |

†The single measured input is $m_e$. The three remaining former inputs — $\alpha$, $m_\mu$, and $\Lambda_{\text{tube}}$ — are derivable from the topological integers (Paper 3).

5. The $(3, 4, 5)$ triple at $k = 3$ explains proton stability, while the absence of any triple at $k = 2$ explains universal meson instability — from the same equation.

6. The pion mass emerges as $E_1/k = 139.6$ MeV, independently of the 23-parameter static fit.

7. All seven decay chains close at $\Delta E/M = 0.0\%$, conserving energy through multiple sector crossings.

8. The three-timescale decay hierarchy (strong/EM/weak) maps directly to three types of sector crossing.

9. The proton mass (945.7 MeV, 0.8%) follows from the same $\alpha_s = 16\alpha$ that predicts $\sin^2\theta_W$ and the same pion mode — no new parameters.

**Testable predictions.** (i) $m_\tau = 1776.9$ MeV at FCC-ee precision (0.03 MeV) [8]. (ii) Normal neutrino mass ordering at JUNO [9]. (iii) $\delta^\nu_{CP} = \pi$ at Hyper-Kamiokande. (iv) $\Sigma m_\nu = 59$ meV, near the cosmological floor [10]. (v) Geometric harmonics at $n \times 139.6$ MeV on the $k = 2$ torus. (vi) Pentaquark narrow widths from $(5, 12, 13)$ exact Pythagorean resonance.

**Code availability.** All numerical calculations and verification code are available at https://github.com/JimGalasyn/null-worldtube.

**Known limitations.** (i) The proton mass derivation uses two variational coefficients — 3/2 for kinetic energy and 2 for the Coulomb factor — not yet derived from NWT dynamics. (ii) The mode-counting conjecture (13 modes) lacks a rigorous spectral-geometry derivation. (iii) The defect-lifetime relationship correctly orders the hierarchy but does not yet yield quantitative lifetimes. (iv) Some mode assignments at similar energies have ambiguity. (v) The quark hierarchy parameter $B^2$ is approximate at ∼3%. (vi) The $V_{cb}$ prediction (5.1%) is the least precise CKM element. These define the program for future work.

## 14 Conclusion

The Standard Model has 26 parameters. We have shown that 23 of them follow from Maxwell's equations on a $(2, 1)$ torus knot, given one measured input ($m_e$) and three topological integers ($p = 2$, $q = 1$, $k = R/r = 3$). The three quantities previously treated as inputs — $\alpha$, $m_\mu$, and $\Lambda_{\text{tube}}$ — are now derivable from the integers alone (Paper 3). Every coefficient is a function of the three integers. The median prediction error is 0.7% across quantities spanning fourteen orders of magnitude, compared to precision measurements [11, 1].

The same geometry determines the dynamics. The Pythagorean resonance condition $(kp)^2 + q^2 = N^2$ classifies all composite particles by aspect ratio, explains proton stability through

the $(3, 4, 5)$ triple, and accounts for universal meson instability through the absence of any fundamental triple at $k = 2$. A four-sector taxonomy — TM, TE, geometric, and open — generates the three-timescale decay hierarchy from three types of sector crossing. Complete decay chains for seven representative mesons conserve energy at every step. The framework does not merely assign parameters — it predicts which particles are stable, which decay channels are available, and how energy flows through the cascade to stable endpoints.

The question "why these parameters?" reduces to "why this knot?" — and the $(2, 1)$ torus knot is the simplest closed curve that produces spin-$\frac{1}{2}$ and breaks $p \leftrightarrow q$ symmetry. The aspect ratio $k = R/r = 3$ is the lowest value admitting a Pythagorean resonance $(3, 4, 5)$, making the proton the unique stable baryon. There is nothing to tune. The Standard Model is not a collection of arbitrary constants. It is the unique electromagnetic and geometric spectrum of a $(2, 1)$ torus knot with aspect ratio $k = 3$.

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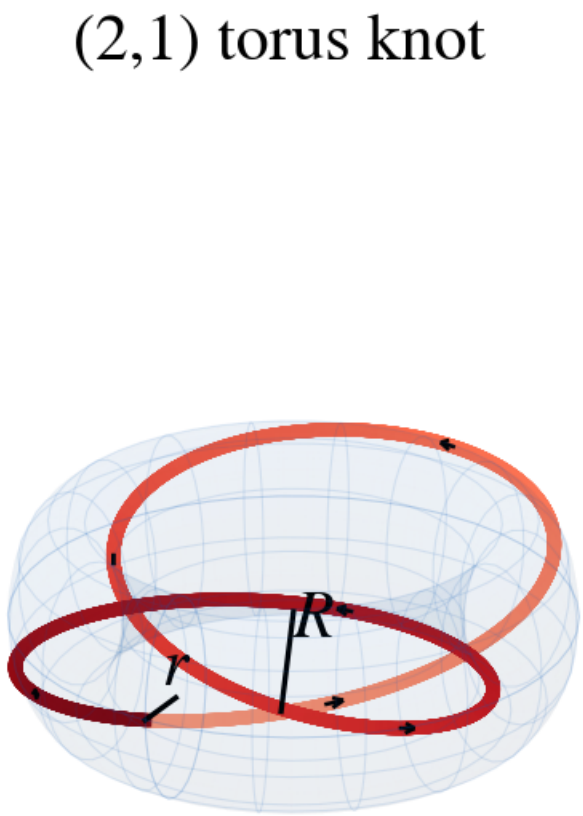


$$\alpha = r/R$$

Figure 2: The $(2, 1)$ torus knot on a transparent torus surface. A photon circulates twice toroidally and once poloidally, requiring $4\pi$ rotation to close — the geometric origin of spin-$\frac{1}{2}$. $R$ is the major radius, $r$ the minor radius, and $\alpha = r/R$ the fine-structure constant. Arrows indicate the current direction $I = ec/L$.

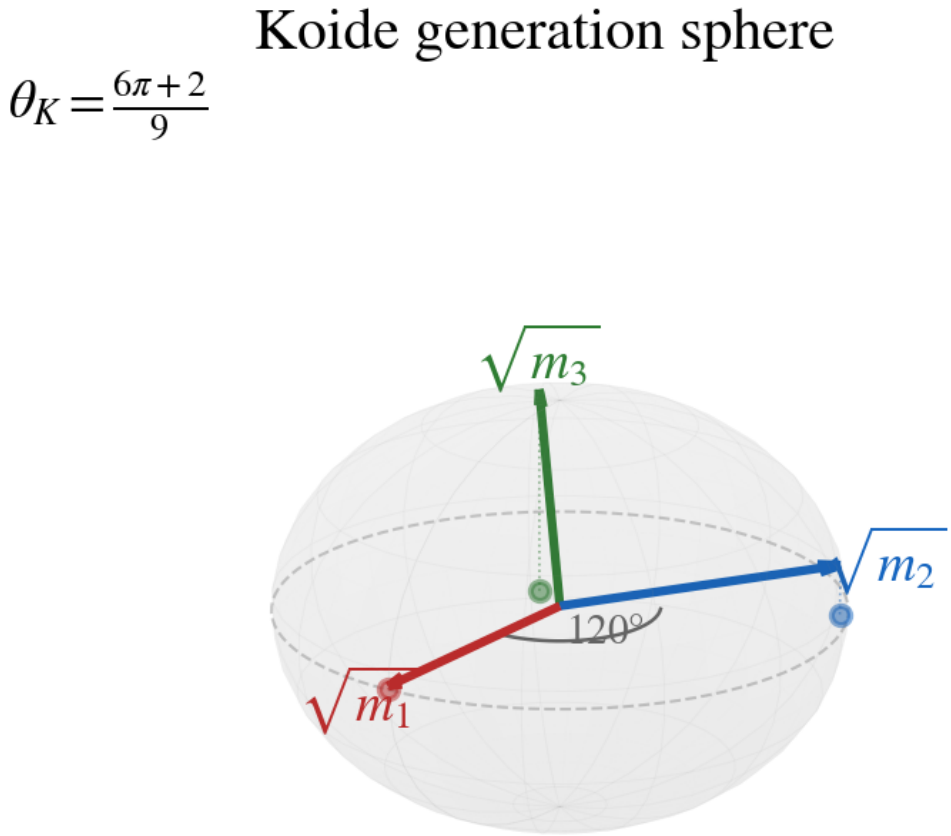


Figure 3: Koide generation sphere. Three mass eigenstates $\sqrt{m_i}$ are represented as unit vectors with 120° azimuthal separation ($\mathbb{Z}_3$ symmetry). The Koide angle $\theta_K = (6\pi + 2)/9$ determines the polar angles. The equatorial projections show the resulting mass ratios.

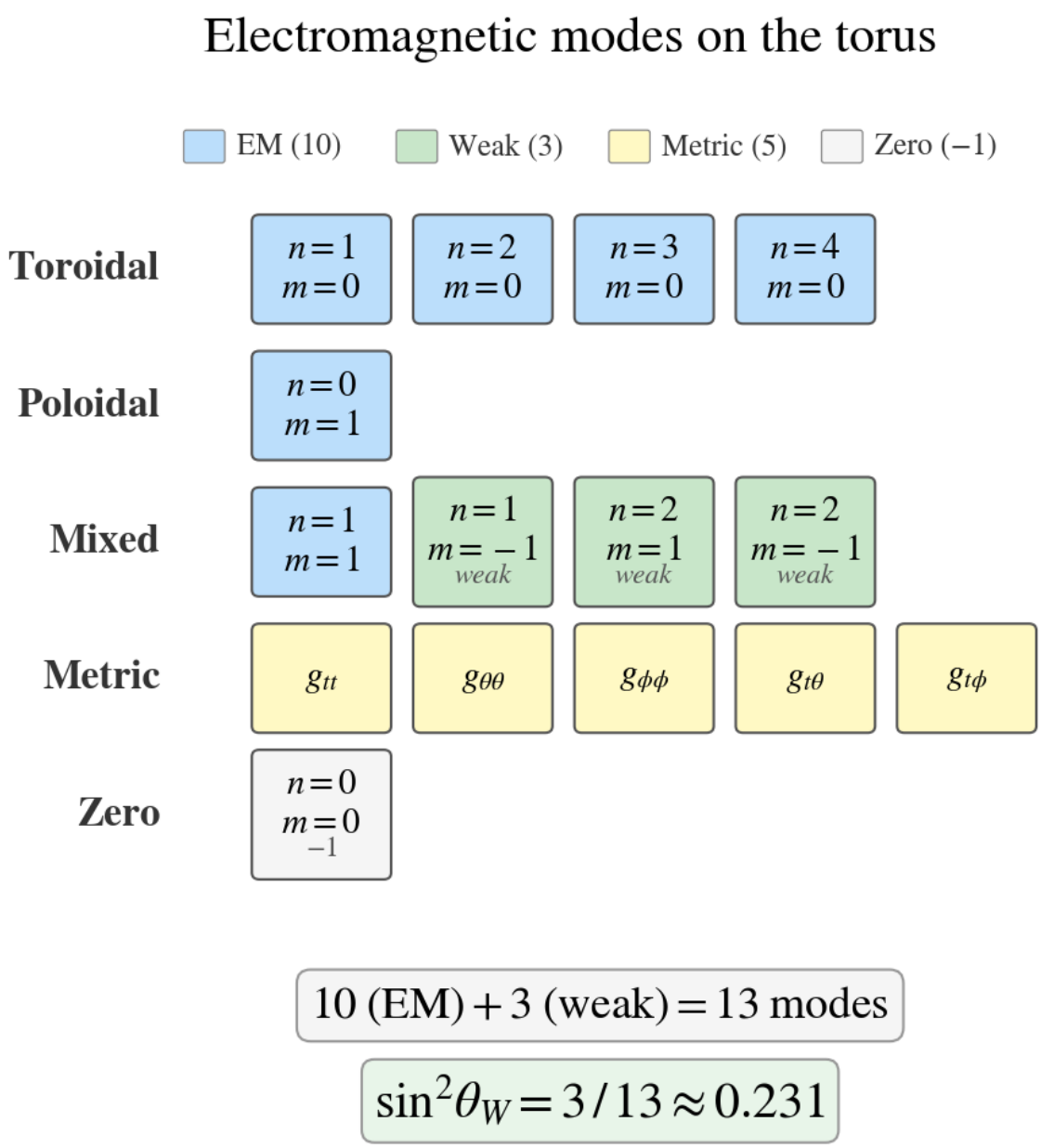


Figure 4: The 13 electromagnetic modes on the $(2, 1)$ torus. Blue: 10 EM-coupled modes (4 toroidal, 1 poloidal, 1 mixed, 4 metric, minus the zero mode). Green: 3 weak-coupled mixed modes. The ratio of weak to total modes gives the Weinberg angle: $\sin^2\theta_W = 3/13 \approx 0.231$.

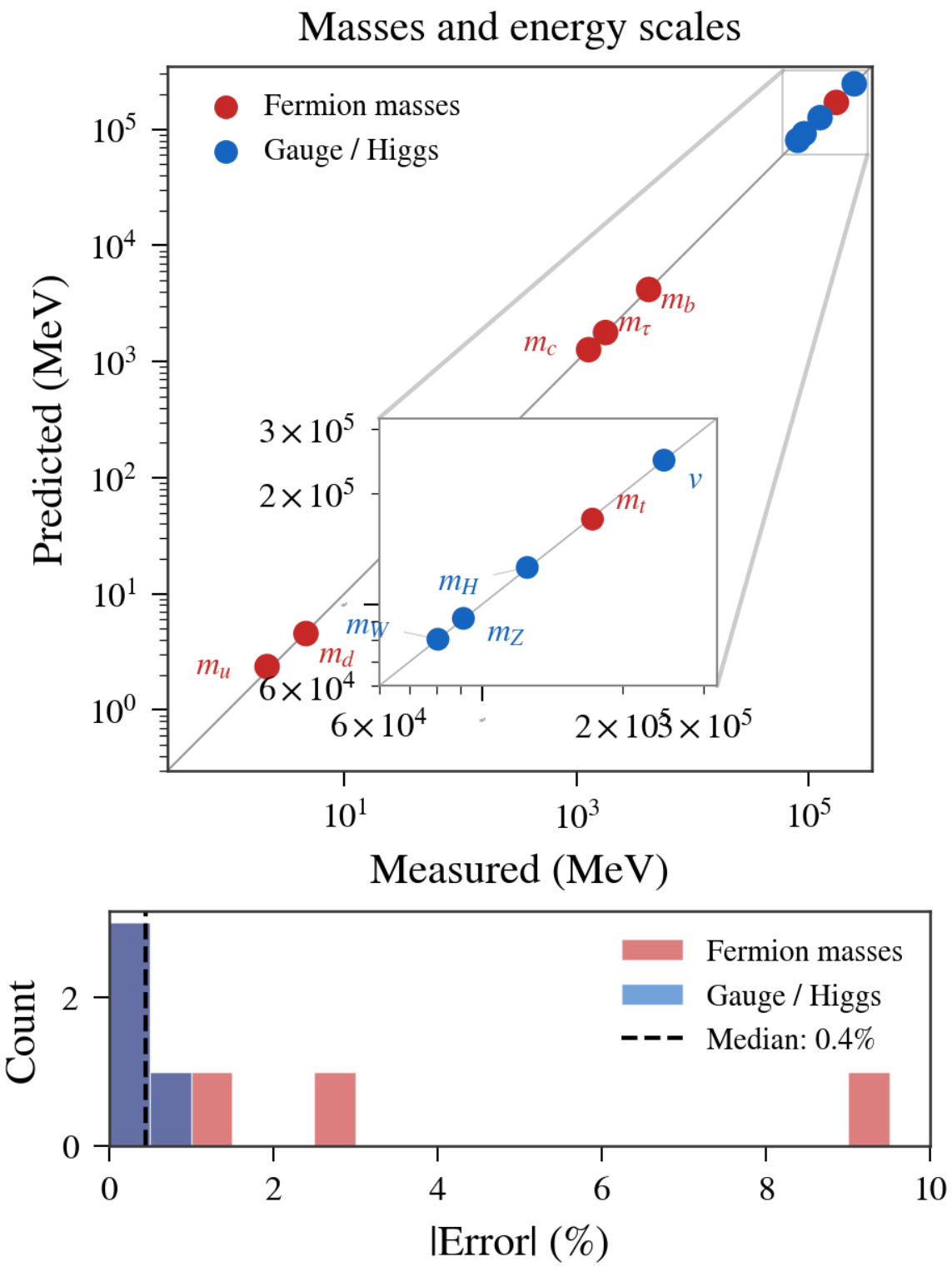


Figure 5: Predicted vs. measured masses and energy scales. Main panel: log-log scatter spanning $m_u \approx 2$ MeV to $v \approx 246$ GeV. Inset: the electroweak cluster ($m_W$, $m_Z$, $m_H$, $m_t$, $v$) resolved in GeV. Bottom: error distribution for the mass sector (median 0.4%).

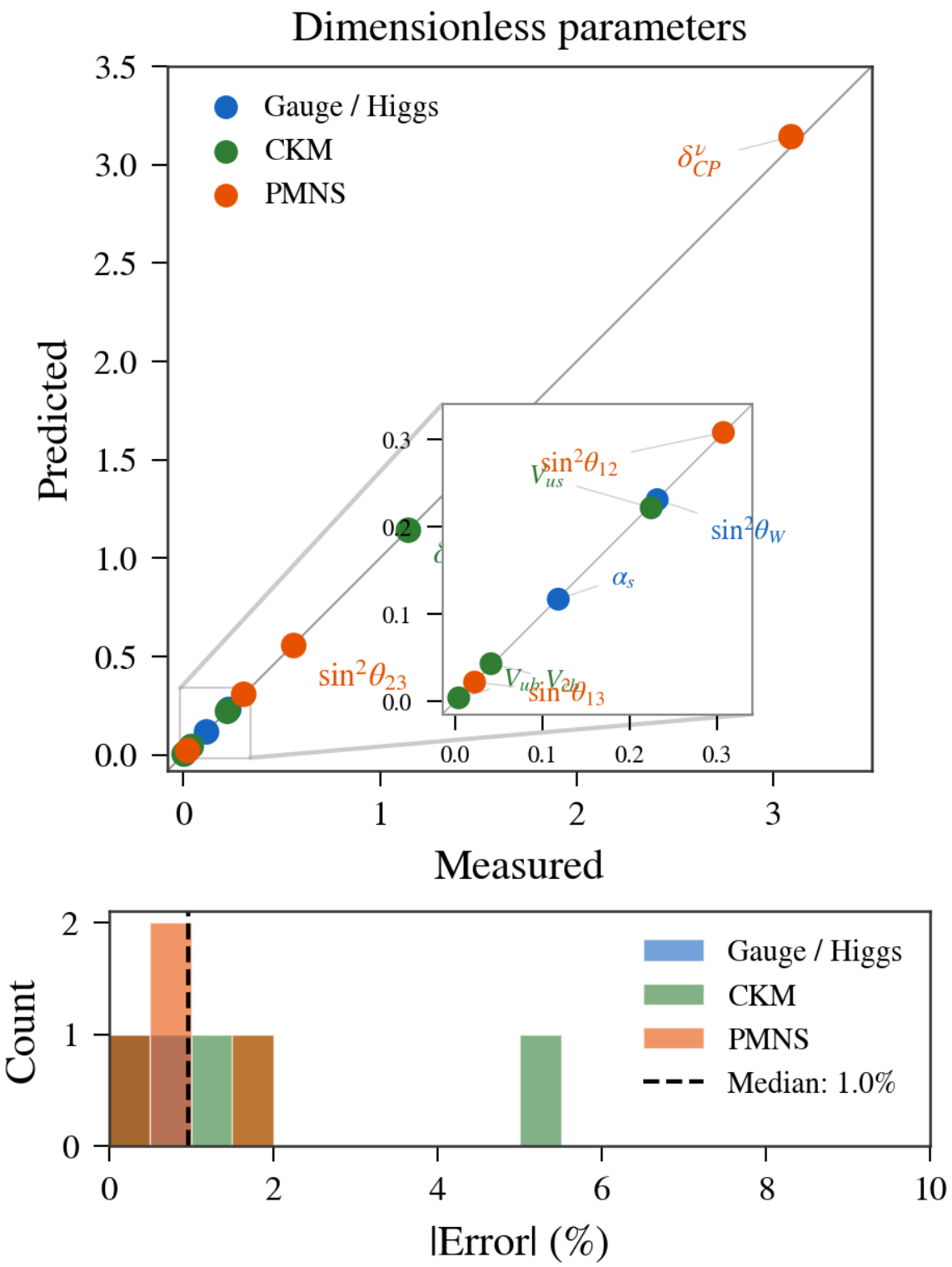


Figure 6: Predicted vs. measured dimensionless parameters. Main panel: linear scatter from $V_{ub} \approx 0.004$ to $\delta^{\nu}_{CP} = \pi$. Inset: the small-value cluster resolved ($V_{ub}$, $\sin^2\theta_{13}$, $V_{cb}$, $\alpha_s$, $V_{us}$, $\sin^2\theta_W$, $\sin^2\theta_{12}$). Bottom: error distribution for dimensionless parameters (median 1.0%).